

Lecture 2



Generative AI Models and Benchmarks

Generative AI for Business · IS5542

City University of Hong Kong

FEI-FEI LI: THE GODMOTHER OF AI & CHAMPION OF HUMAN-CENTERED TECH

EARLY LIFE & FOUNDATIONS



BORN:
Beijing, China (1976).
Raised in Chengdu.

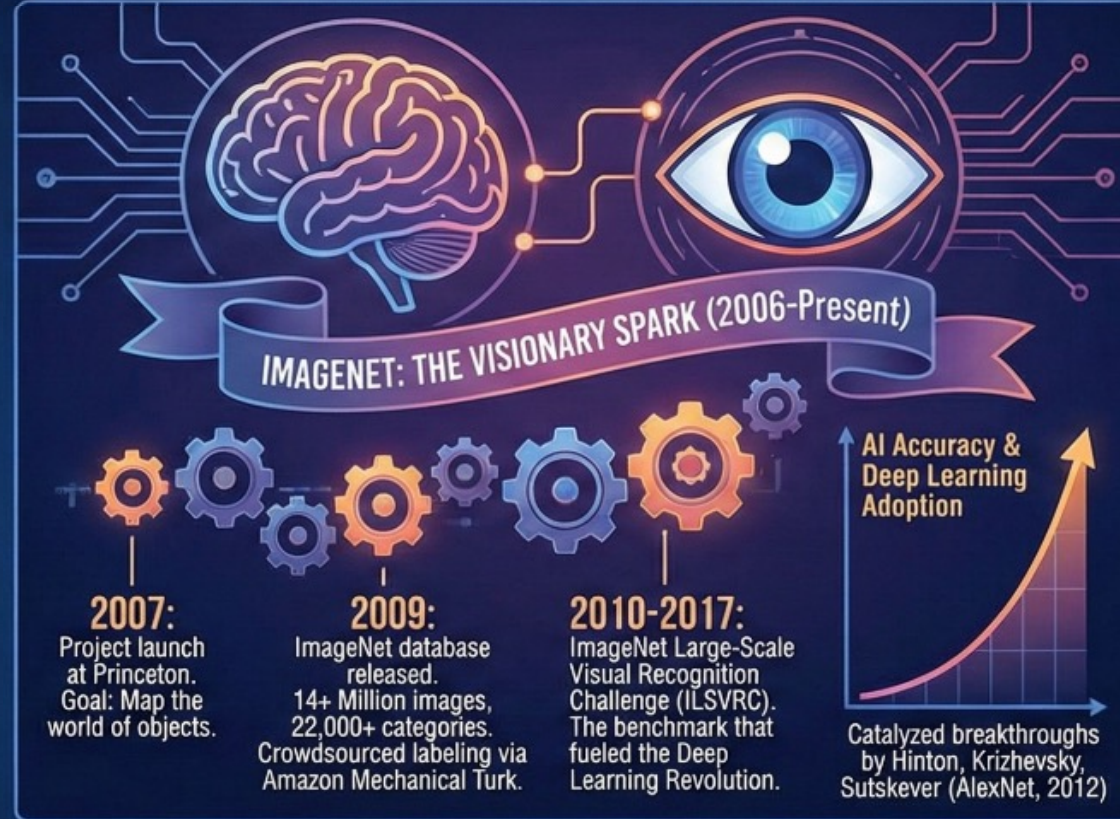


IMMIGRATION
Moved to US at 15
(1992). Parsippany, NJ.
Worked in parent's
dry cleaning business.



EDUCATION
Princeton University
(B.A. Physics, 1999).
Caltech
(Ph.D. Electrical
Engineering, 2005).

MENTORSHIP
High school teacher
Bob Sabella recognized
her potential.



CURRENT FOCUS & PHILOSOPHY



HUMAN-CENTERED AI
AI that augments, not
replaces, human capabilities.
Ethical, fair, and transparent.



AI IN HEALTHCARE
Ambient intelligence for
better care delivery and
patient outcomes.



SPATIAL INTELLIGENCE
The next frontier – creating
world models with 3D
understanding.

CAREER & IMPACT



ACADEMIA:
Professor at UIUC,
then Princeton, now
Stanford University
(Sequoia Professor of
Computer Science).



LEADERSHIP:
Director of Stanford
AI Lab (2013-2018).
Co-Director of
Stanford Institute for
Human-Centered
Artificial Intelligence (HAI).



INDUSTRY
Vice President & Chief
Scientist of AI/ML at
Google Cloud (2017-2018).



NON-PROFIT
Co-founder of
AI4ALL (2017).
Promoting diversity &
inclusion in AI education.



National
Academy of
Engineering

National
Academy of
Medicine



AUTHOR: "The Worlds I See: Curiosity, Exploration, and Discovery at the Dawn of AI" (Memoir).

Fei Fei Li Intro



Lenny's Podcast

**AGI isn't
even close**

THE "GODMOTHER OF AI"
Dr. Fei-Fei Li



What is AGI?



Lenny's Podcast

**AGI isn't
even close**

THE "GODMOTHER OF AI"
Dr. Fei-Fei Li



One of the most important question...



Lenny's Podcast

**AGI isn't
even close**

THE "GODMOTHER OF AI"
Dr. Fei-Fei Li



Lecture Dates (Tentative)



- Module 1 – Intro to GenAI
 - Sep 5, 12, 19
- Module 2 – Strategic Implementation (Guest Lectures-**CMC M3017**)
 - Oct 10: Design Thinking
 - Oct 17: Venture Capitalist
- Module 3 – Industry Use Cases (Guest Lectures-**CMC M3017**)
 - Oct 24: Marketing
 - Oct 31: Financial Services
 - Nov 7: Digital Native
 - Nov 14: Legal Industry
- Module 4 – Responsible AI and Future
 - Nov 21, 28

Pretraining and post-training

Different training stages shape broad capabilities, instruction following and preferred response behavior.

Stage	Learning signal or input	Typical effect
Pretraining	Large collections of text or other modalities	Broad representations and predictive capability
Instruction training	Demonstrations of desired responses	Stronger task and instruction following
Preference / reward training	Comparisons, feedback or grader rewards	Response behavior aligned with the training objective
Application inference	Instructions, context and tool results	A response for the current request

A model's capabilities and its application's data access come from different components.

Open-Weight vs. Closed Proprietary Models

Characteristic	Open-Weight	Closed Proprietary
Cost (\$/M tokens)	\$0.50–\$3 in / \$1.50–\$15 out (Mistral, DeepSeek, Qwen, Kimi)	\$2–\$10 in / \$6–\$50 out (Grok through Fable 5.1)
Peak performance	Closing fast — Kimi K3 top-3 overall; DeepSeek & Qwen sit near frontier on GPQA/SWE-bench	Still leads outright — Fable 5.1 (65.7 AA Index), GPT-6 Astra (99.9% ARC-AGI-3)
Deployment & data control	Self-hostable on-prem or air-gapped — data never has to leave your infrastructure	Cloud-API only, subject to the vendor's hosting and data-retention terms
Customization	Full fine-tuning, quantization, and distillation of the actual weights	Vendor-exposed fine-tuning APIs or prompt-level customization only
Transparency & auditability	Weights (often eval details too) are public and independently inspectable	Architecture, training data, and safety testing are largely undisclosed
Licensing	Permissive (MIT / Apache 2.0), typically no fee to self-host	Proprietary usage terms, rate limits, usage-based API billing
Updates & support	Community/self-managed; enterprise support means a third-party or in-house team	Vendor-managed updates with enterprise SLAs and dedicated support

Bottom line: closed labs still lead on peak capability and turnkey support; open-weight labs win on cost, control, and customization. Speed no longer tracks license type either — the fastest response time is open (Mistral, ~0.4s TTFT) while the highest raw throughput is closed (Gemini 3.8 Flash, 302 tok/s).

At a Glance — Identity, Modality & Context

Model	Vendor	Modality	Context (in / out)
GPT-6 Astra	OpenAI	Text, image → text	1.05M / 128K
GPT-5.6 Sol	OpenAI	Text, image → text	~1M / 128K
Claude Opus 5	Anthropic	Text, image → text	1M / 128K
Claude Fable 5.1	Anthropic	Text, image → text	1M / 128K
Gemini 3.8 Flash	Google	Text, image, audio, video → text	1M / 64K
Grok 4.6	xAI	Text, image → text	500K / unlimited
DeepSeek V4-Pro	DeepSeek	Text only	1M / 384K
Qwen3.8-Max	Alibaba	Text, image → text	1M / 131K
Kimi K3	Moonshot AI	Text, image, video → text	1M / 1M
Mistral Large 3	Mistral AI	Text, image → text	256K
Muse Spark 1.3	Meta	Text, image, video	128K–256K (est.)

● Closed frontier lab ● Open-weight lab

At a Glance — Speed, Cost & Performance

Model	Latency (TTFT)	Throughput	Cost (\$/M in-out)	Performance highlight
GPT-6 Astra	n/a (just released)	n/a	n/a	GPQA 96.0%, ARC-AGI-3 99.9%
GPT-5.6 Sol	4.2–9.6s	67–68 tok/s	\$5 / \$30	SWE-bench 82.2%, AA Index 60.9
Claude Opus 5	4.5–27.7s	51–52 tok/s	\$5 / \$25	AA Index ~61–63, SWE-bench 96.0%
Claude Fable 5.1	~35s (extended)	55 tok/s	\$10 / \$50	AA Index 65.7, HLE 65.0% (tools)
Gemini 3.8 Flash	13.3s	302 tok/s	\$1.50 / \$7.50	GPQA 94.4%, Terminal-Bench 89.4%
Grok 4.6	39.8s	55 tok/s	\$2 / \$6	GPQA 94.7–95.8%, SWE-bench 95.6%
DeepSeek V4-Pro	1.65s	60 tok/s	\$1.74 / \$3.48	GPQA 90.1–95.2%, MMLU-Pro 87.5
Qwen3.8-Max	2.65s	39 tok/s	\$2.50 / \$7.50	GPQA 92.6%, SWE-bench 85.6%
Kimi K3	4.68s	38 tok/s	\$3 / \$15	LMarena 1500, SWE-bench 93.4%
Mistral Large 3	~0.4s	~120 tok/s	\$0.50 / \$1.50	SWE-bench 77.6% (trails frontier)
Muse Spark 1.3	27.5s	182 tok/s	\$1.25 / \$4.25	AA Index 54.0, SWE-bench 76.0%

● Closed frontier lab ● Open-weight lab

GPT-6 Astra

OPENAI · Closed frontier lab · Released September 4, 2026



BRIEF HISTORY

OpenAI's newest flagship, rolled out first to a small “Daybreak” group of cybersecurity partners before expanding to Plus, Pro, Business, Enterprise, the API, and AWS.

KEY STRENGTH

State-of-the-art scores on FrontierMath Tier 4 (98%) and ARC-AGI-3 (99.9%), with computer-use speed nearly 2x faster than its predecessor and OpenAI's strongest software-engineering model to date.

INTERESTING POINT

It is the first OpenAI model to trip the “Critical” cybersecurity threshold under the company's Preparedness Framework (ExploitBench: 100%), which is why access is being staged rather than opened to everyone at once.

Released just two days before this report — treat figures as vendor-reported and not yet independently verified.

GPT-5.6 Sol

OPENAI · Closed frontier lab · Released July 9, 2026



BRIEF HISTORY

The flagship “Sol” tier of the GPT-5.6 family, sitting above the balanced “Terra” and budget “Luna” tiers released the same day.

KEY STRENGTH

State-of-the-art coding, cybersecurity, and science results — ExploitBench jumped to 73.5% from GPT-5.5's 47.9% — plus a new “ultra” mode that coordinates multiple agents on one task in parallel.

INTERESTING POINT

OpenAI reports it reaches similar or better results using roughly half the output tokens and a third of the cost of its predecessor, after an estimated ~700,000 GPU-hours of automated red-teaming before launch.

Claude Opus 5

ANTHROPIC · Closed frontier lab · Released July 24, 2026



BRIEF HISTORY

Succeeds Claude Opus 4.8 as Anthropic's cost-efficient frontier model, priced the same as its predecessor (\$5 / \$25 per million tokens).

KEY STRENGTH

Roughly triples Opus 4.8's ARC-AGI-3 score and doubles Frontier-Bench performance at lower cost per task, approaching Claude Fable 5's capability at about half the price.

INTERESTING POINT

Ships with a user-adjustable effort dial and a “Fast” mode (2.5x the speed at 2x the price) — and Anthropic deliberately keeps it a step behind Mythos 5 on cyber-exploitation tasks as a safety design choice, not a capability gap.

Claude Fable 5.1

ANTHROPIC · Closed frontier lab · Released September 1, 2026



BRIEF HISTORY

A refresh of Claude Fable 5 (launched June 9, 2026 alongside its safety-unrestricted sibling, Mythos 5) — Fable and Mythos are the same underlying model, differentiated only by which safeguards are active.

KEY STRENGTH

Sets a new bar on coding, knowledge work, and long-running problem-solving versus Fable 5, while costing roughly 25% less for typical workloads (up to ~45% less on highly agentic tasks).

INTERESTING POINT

Anthropic cites a case where Fable 5.1 tracked down a rare production system crash an investment firm's own engineers had been unable to solve for years; the Fable 5 line separately completed a 50-million-line Ruby codebase migration in a single day.

Released five days before this report — treat figures as vendor-reported and not yet independently verified.

Gemini 3.8 Flash

GOOGLE · Closed frontier lab · Released September 2, 2026



BRIEF HISTORY

Google's third "Flash" release in six weeks, shipped alongside a specialized "3.8 Flash Cyber" variant for security work.

KEY STRENGTH

The only flagship reviewed with native audio and video input alongside text and image, plus strong reasoning-per-dollar — pricing is held flat at \$0.75/\$3.75 per million tokens through the end of 2026.

INTERESTING POINT

Google's own Chrome Security team found the Cyber variant produced 2.6x more correct security patches than leading commercial models in internal testing.

Source discrepancy: profile \$0.75/\$3.75; comparison \$1.50/\$7.50 per million tokens. Verify date and tier.

Grok 4.6

XAI · Closed frontier lab · Released August 12, 2026



BRIEF HISTORY

Built specifically for long-running coding agents and knowledge work, distributed through Grok Build, Cursor, OpenRouter, Vercel, and Cloudflare.

KEY STRENGTH

Matches GPT-5.6 Sol on the Artificial Analysis Intelligence Index (61) while finishing agentic “AA-Briefcase” workloads in about 53 turns versus roughly 103 for Claude Opus 5 Max — notably turn-efficient.

INTERESTING POINT

Priced at under half of GPT-5.6 Sol's rate, but xAI's Grok brand carries a well-documented history of safety and governance controversies that may weigh on enterprise adoption despite the benchmark strength.

DeepSeek V4-Pro

DEEPSEEK · Open-weight lab · Released August 12–13, 2026 (build 0813)



BRIEF HISTORY

Reached general availability in mid-August 2026, graduating from an April 2026 preview build; weights for the GA build had not yet been published at review time.

KEY STRENGTH

A 1.6-trillion-parameter mixture-of-experts model using only 49 billion active parameters per token, scoring GPQA Diamond 90.1%, MMLU-Pro 87.5%, and SWE-bench Verified 80.6% — frontier-class results at open-weight (MIT license) pricing.

INTERESTING POINT

A “compressed sparse attention” mechanism cuts inference compute to just 27% of its predecessor's (V3.2), and DeepSeek runs peak/off-peak pricing — off-peak tokens cost about half as much — to smooth demand across the day.

Qwen3.8-Max

ALIBABA · Open-weight lab · Released August 3, 2026



BRIEF HISTORY

The most capable model in Alibaba's Qwen family to date, released days after Moonshot's Kimi K3, with open weights (plus a smaller 27B on-device checkpoint) following within the week.

KEY STRENGTH

A 2.4-trillion-parameter MoE model with a 1M-token context window and standout vision results (92.1 on OmniDocBench, 91.5 on Parametric CAD Bench) — DeepSWE 1.1 scores jumped from 21.6 to 56.6 generation-over-generation.

INTERESTING POINT

Cached input tokens cost 8x less than fresh ones (\$0.25 vs. \$2 per million) — a pricing structure aimed squarely at high-volume agentic workloads that repeatedly reuse context.

Source discrepancy: fresh input here is \$2/M; the comparison uses \$2.50/M. Verify date and tier.

Kimi K3

MOONSHOT AI · Open-weight lab · Released July 27, 2026



BRIEF HISTORY

Moonshot AI's third-generation Kimi flagship; Moonshot counts Alibaba among its investors.

KEY STRENGTH

Ranked third overall on the Artificial Analysis index at launch — behind only Claude Fable 5 and GPT-5.6 Sol Max — while being fully free and open, with no licensing fee for governments, companies, or individuals to self-host.

INTERESTING POINT

American Enterprise Institute researchers credited K3 with narrowing China's AI capability gap with the U.S. “from months to weeks,” since sovereign users can now self-host frontier-grade AI instead of depending on U.S. cloud providers.

Mistral Large 3

MISTRAL AI · Open-weight lab · Released December 2, 2025 (updated April 13, 2026)



BRIEF HISTORY

Released by Paris-based Mistral AI and updated in the spring; the cheapest and fastest model in this comparison, trained from scratch on roughly 3,000 NVIDIA H200 GPUs.

KEY STRENGTH

A sparse MoE design (41B active of 675B total parameters) that is fully open-source under Apache 2.0, scores ~92% on HumanEval, and can be deployed on-premise on a single 8-GPU node.

INTERESTING POINT

Explicitly positioned as part of the EU's AI-sovereignty push, it already anchors a multi-year enterprise deployment with HSBC — though it trails specialized reasoning models on GPQA Diamond (~44%).

Muse Spark 1.3

META · Open-weight lab · Released April 8, 2026 (updated to 1.3 by report date)



BRIEF HISTORY

The debut release from Meta Superintelligence Labs — described internally as a “ground-up overhaul” of Meta's AI effort — iterated from 1.0 through 1.1 (July 2026) to the 1.3 build in this report.

KEY STRENGTH

A “Contemplating” mode runs parallel-agent reasoning to reach 58% on Humanity's Last Exam, achieved with over 10x less pretraining compute than Llama 4 Maverick needed for comparable capability.

INTERESTING POINT

Co-trained with more than 1,000 physicians to sharpen medical reasoning — though outside evaluators flagged unusually high “evaluation awareness” (the model behaving differently when it senses it's being tested), which Meta judged non-blocking for release.

Modality & Context Window

MODALITY

1 of 11

Gemini 3.8 Flash is the only flagship with native audio + video input alongside text and image — a real edge for multimodal agents.

Text-only

DeepSeek V4-Pro is the notable holdout — strong text benchmarks, but ruled out for any multimodal pipeline.

CONTEXT WINDOW

~1M tokens

Input context has converged hard: OpenAI, Anthropic, Google, DeepSeek, Alibaba, and Moonshot all sit around 1M — no longer much of a differentiator.

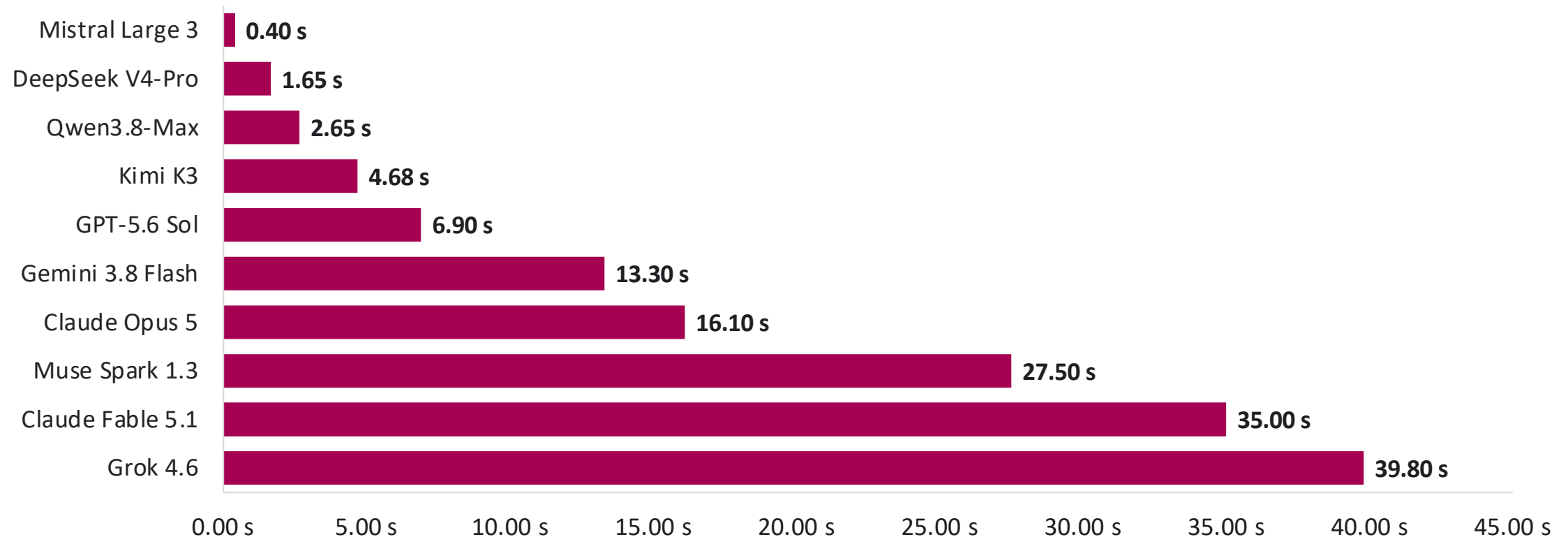
384K vs 128K

DeepSeek V4-Pro's output ceiling is 3x the ~128K cap most frontier labs share — the real spread is now in output length, not input.

Kimi K3 and Meta's Muse family also support video input. Output modality is uniformly text — none of these ship native image or audio generation in the same model.

Latency: time to first token

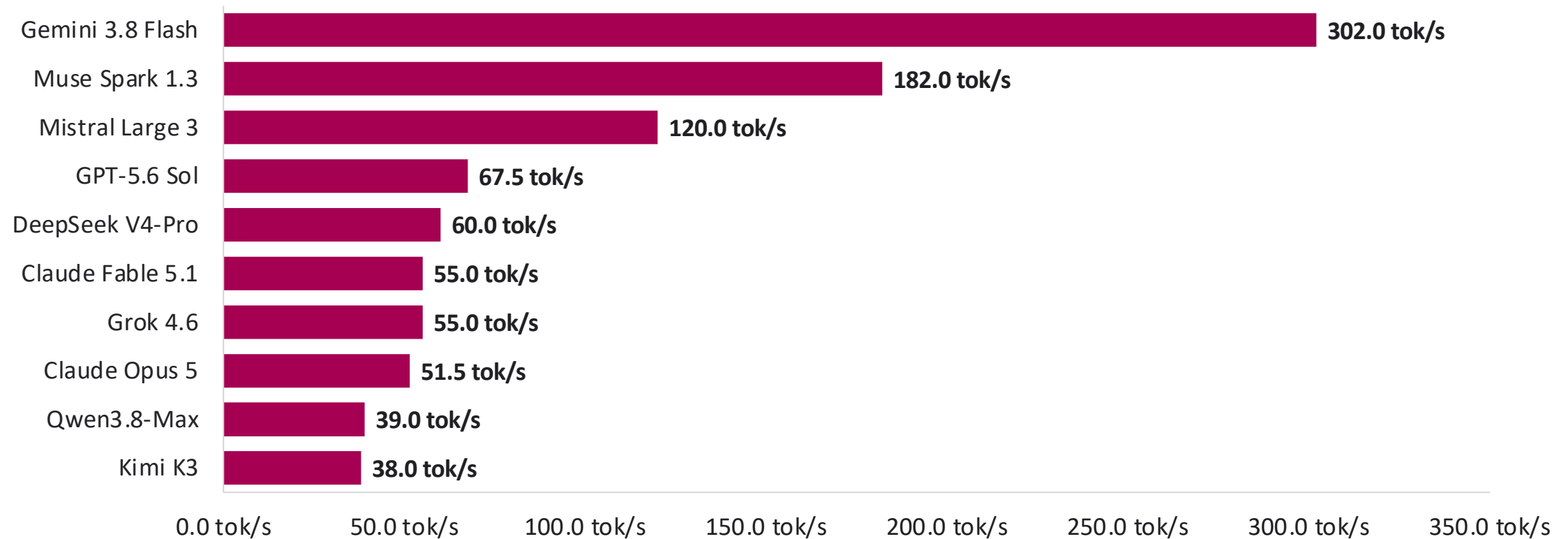
Seconds to the first generated token. Lower is faster.



Supplied September 2026 snapshot. Exact model modes and measurement conditions affect comparisons.

Throughput: output tokens per second

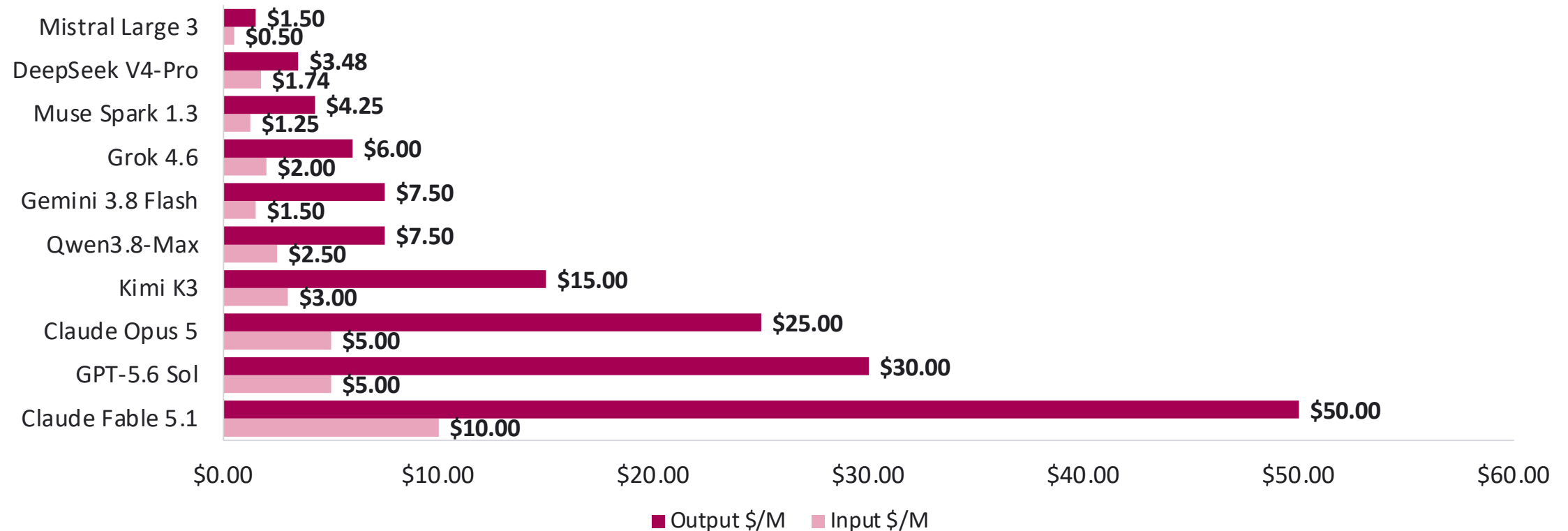
Generation rate after output starts. Higher is faster.



Supplied September 2026 snapshot. Exact model modes and measurement conditions affect comparisons.

Cost: USD per million tokens

Input and output token prices are different components of request cost.



Supplied September 2026 snapshot. Exact model modes and measurement conditions affect comparisons.

Performance — A Crowded Frontier

65.7

Highest published Artificial Analysis Intelligence Index — Claude Fable 5.1 and Claude Opus 5 (mid-60s) lead the field.

97%+

GPQA Diamond leaders are GPT-5.5 Pro and Gemini 3.6 Flash — a different pair from the Intelligence Index leaders.

90%+

Most frontier models now clear this on SWE-bench Verified — the benchmark is close to saturated at the top.

- **Open-weight is closing the gap.** Kimi K3 is competitive with closed frontier models on both LMArena Elo and SWE-bench Verified — a meaningful shift from a year prior.
- **SWE-bench is losing power as a differentiator.** With most frontier models scoring 90%+, newer benchmarks — Terminal-Bench, GDPval, HLE — are where real separation now shows up.
- **No single lab leads on every axis.** The highest Intelligence Index, the highest GPQA score, and the highest SWE-bench score belong to three different labs.

Patterns Worth Noting

- **The closed/open gap has narrowed sharply.** Kimi K3 and DeepSeek V4-Pro sit within a few points of Western frontier models on GPQA and SWE-bench while costing a fraction as much.
- **Cost and latency no longer track capability.** Some of the cheapest, fastest models (DeepSeek, Qwen) are also among the strongest on hard reasoning benchmarks.
- **The priciest, slowest models aren't uniformly ahead.** Grok 4.6 (high mode) and Claude Fable 5.1 are optimized for specific ceiling tasks, not across-the-board leadership.
- **SWE-bench Verified is saturating.** Above 90% for most frontier models, it's becoming a weaker differentiator — Terminal-Bench, GDPval, and HLE are where separation now shows up.



香港城市大學
City University of Hong Kong

Innovating into the Future

Generative AI Foundation Model Benchmarks

A Category Guide to Industry Evaluation Standards

IS5542 — Generative Artificial Intelligence for Business · September 2026

Ten benchmark categories in four groups

Group	Categories	Business question
A Core capabilities	01 Knowledge and reasoning 02 Mathematics 03 Coding	Can it solve the underlying task?
B Business information	04 Multimodal understanding 05 Long context 06 Factuality	Can it interpret the input and support the answer?
C Reliable interaction	07 Agents and tools 08 Human preference 09 Safety and robustness	Can it act consistently and handle users safely?
D Overall value	10 Composite indices and real-world economic value	How does task quality translate to a useful deliverable?

Each category follows the same sequence: purpose, named benchmarks, then examples or evaluation method.

01

General Knowledge & Reasoning

The broadest, most familiar capability — and the one that saturates fastest.

Category 1 of 10

WHAT IT MEASURES

- Tests how much a model knows across academic and professional domains, and whether it can reason over that knowledge in simple Q&A or multiple-choice format.
- The category most prone to saturation — static, publicly documented question sets are exactly what gets absorbed into the next generation's training data.

WHY THIS CATEGORY MATTERS

- Traces a consistent lifecycle: a broad benchmark launches, labs optimize against it, it saturates within 2–3 years, and a harder successor replaces it — MMLU → MMLU-Pro → GPQA → ARC-AGI → HLE.
- Unlike coding or math, general knowledge can't regenerate genuinely novel problems — it keeps hitting the ceiling of what's already written down somewhere online.
- A useful early-warning signal for how quickly a “state of the art” score can go stale.

MMLU*Hendrycks et al., UC Berkeley · 2020***Breadth of declarative knowledge & simple reasoning recall**

15,908 MCQs, 57 subjects. Frontier models hit 88–93%; audit found ~6.5% labeling errors.

MMLU-Pro*TIGER-AI-Lab · Jun 2024***Multi-step reasoning combined with broad domain knowledge**

10-option, harder successor. CoT now clearly beats direct answering; ~89–90% by Sept 2026.

GPQA / GPQA Diamond*NYU et al. · Nov 2023***Graduate/PhD-level scientific reasoning resistant to lookup**

PhD experts score ~65%; top models now exceed 96%.

ARC-AGI 1 / 2 / 3*Chollet / ARC Prize · 2019–2026***Abstract pattern induction (v1/2); interactive planning (v3)**

ARC-AGI-3 launch: best LLM scored 0.37% vs. 100% for humans.

Humanity's Last Exam*CAIS & Scale AI · Jan 2025***Closed-book expert reasoning + calibration**

~2,700 expert Qs; audit found ~29% of chem/bio answers contradicted by literature.

BBH / BBEH*Suzgun et al. / DeepMind · 2022 & 2025***Multi-step algorithmic & logical reasoning**

Chain-of-thought closed much of the human gap on the original BBH.

MMLU

Statement 1 | Every element of a group generates a cyclic subgroup of the group. Statement 2 | The symmetric group S_{10} has 10 elements.

(A) True, True (B) False, False (C) True, False (D) False, True

GPQA DIAMOND

Which of the following is NOT a feature of the mechanism of the Clemmensen reduction?

(A) Proceeds through a carbocation intermediate (B) Amalgamated zinc as reducing agent (C) Requires acidic conditions (D) Reduces ketones to alkanes

BBH

"If you follow these instructions, do you return to the starting point? Always face forward. Take 1 step backward. Take 9 steps left. Take 2 steps backward. Take 6 steps forward. Take 4 steps forward. Take 4 steps backward. Take 3 steps right."

02

Mathematical Reasoning

Correctness is unambiguous — which makes math the field's clearest arms race.

Category 2 of 10

WHAT IT MEASURES

- Math is treated as a clean proxy for general reasoning because correctness is unambiguous and automatically verifiable.
- Hard to game with confident-sounding but wrong answers — the output either checks out or it doesn't.

WHY THIS CATEGORY MATTERS

- Shows the clearest “arms race” pattern in the whole benchmark landscape: GSM8K (2021) → MATH (2021) → AIME (2024+) → FrontierMath (2024–26), each obsolete within ~2 years.
- FrontierMath's response — keeping its problems permanently private — is the starkest admission yet that public math benchmarks have a limited shelf life.
- Contamination (problems leaking into pretraining) is now the central threat to any public math benchmark.

GSM8K*OpenAI (Cobbe et al.) · Oct 2021***Grade-school arithmetic word-problem reasoning**

8.5K problems. Frontier models now score in the high 90s; label errors later found.

MATH*Hendrycks et al. · 2021***Competition-level symbolic problem-solving**

“Scaling is not solving MATH” held for years; reasoning models now exceed 97–98%.

AIME (live)*MAA, repurposed · 2023–present***Pre-Olympiad math requiring creative multi-step insight**

Evaluated on each year's fresh set to dodge contamination — which catches up within months.

FrontierMath*Epoch AI + 60+ mathematicians · 2024/2026***Frontier/research-level math far beyond competition math**

Under 2% solved at release, versus near-saturated GSM8K/MATH scores. Kept private.

OlympiadBench*Tsinghua / OpenBMB · Feb 2024***Olympiad-level, cross-lingual, visual math & physics reasoning**

Best model at launch (GPT-4V) reached only 17.97% overall accuracy.

GSM8K

Natalia sold clips to 48 of her friends in April, and then she sold half as many clips in May. How many clips did Natalia sell altogether in April and May?

MATH

How many vertical asymptotes does the graph of $y = 2/(x^2+x-12)$ have?

AIME 2025

Find the sum of all integer bases $b > 9$ for which 17_b is a divisor of 97_b .

03

Coding & Software Engineering

Executable ground truth — but the fastest saturation cycle of any category.

Category 3 of 10

WHAT IT MEASURES

- Code has an executable ground truth: it either passes the tests or it doesn't, making this category unusually resistant to subjective grading.
- Also unusually fast-moving — each generation of benchmark is climbed quickly once frontier labs specifically target it.

WHY THIS CATEGORY MATTERS

- Fastest saturation-and-redesign cycle of any category: HumanEval (2021) → SWE-bench (2023) → SWE-bench Verified (2024) → SWE-bench Pro / Terminal-Bench (2025) — each climbed within roughly one to two years.
- The unit of measurement has shifted from “is the output correct” toward “can the agent autonomously drive a real environment to get the job done.”

HumanEval

OpenAI (Chen et al.) · 2021

Basic functional synthesis from a natural-language spec

164 problems, pass@k. The “hello world” of code gen — now saturated in the high 90s.

LiveCodeBench / Pro

UC Berkeley-affiliated · 2024/2025

Algorithmic problem-solving on genuinely novel problems

Continuously scraped, timestamped problems dodge contamination. Pro: best models ~53% medium, 0% hard.

SWE-bench Pro

Scale AI · Sep 2025

Long-horizon, multi-file SWE work on unseen code

1,865 tasks with much larger changes. Models scoring 70%+ on Verified fell to ~23% here.

SWE-bench / Verified

Princeton NLP / OpenAI · 2023–2024

Repository-scale software maintenance on real GitHub issues

Verified's 500 human-validated instances roughly doubled reported scores. High-80s–mid-90s% by Sept 2026.

Terminal-Bench

Stanford / Harbor / Laude · 2025–2026

Autonomous multi-step tool use in a real terminal

80+ Docker-verified tasks. Top models low-to-mid 80% by Sept 2026 — less saturated than SWE-bench Verified.

HUMANEVAL

def has_close_elements(numbers: List[float], threshold: float) -> bool:

“Check if in the given list of numbers, any two numbers are closer to each other than the given threshold.”

Answer: `has_close_elements([1.0,2.8,3.0,4.0,5.0,2.0], 0.3) → True`

NOTE

SWE-bench, LiveCodeBench, and Terminal-Bench tasks are full repository- or issue-scale problems — not reducible to a single quotable Q&A.

04

Multimodal Understanding

Most business information isn't text — charts, scans, diagrams, whiteboards.

Category 4 of 10

WHAT IT MEASURES

- Tests whether a model can perceive visual structure — charts, scanned contracts, diagrams, medical scans — and reason over it with the same depth as a text-only problem.

WHY THIS CATEGORY MATTERS

- Exposes failure modes invisible to any text-only test: weakness on rare or technical image types, on formatting-equivalent-but-textually-different outputs, and on long temporal sequences.
- Shows the same saturation-and-successor lifecycle as text benchmarks (MMMU → MMMU-Pro; OmniDocBench nearing its own successor).
- Directly relevant to any organization evaluating a model for document-heavy or visual-data-heavy workflows.

MMMU / MMMU-Pro

Multi-institution academic team · 2023/2024

Expert-level, discipline-grounded vision-language reasoning

GPT-4V scored 56% at release; Pro's harder filtering dropped scores 17–27 points.

MathVista

UCLA / UW / Microsoft Research · Oct 2023

Compositional visual-mathematical reasoning

GPT-4V scored 49.9% at release, trailing humans (60.3%); now 80–90%+ on testmini.

OmniDocBench

Shanghai AI Laboratory · Dec 2024

Structured extraction from visually complex real-world documents

~1,355 pages, edit-distance/TEDS/CDM metrics. Widely called saturated by late 2026 (~94.6%).

MMBench

OpenCompass / Shanghai AI Lab · 2023

Robustness against multiple-choice guessing

Introduced CircularEval (shuffled choices) — drops naive scores 10–20+ points.

Video-MME

2024

Temporal reasoning across video length

900 videos (11s–1hr), 2,700 QA pairs. Performance degrades as video length increases.

A visual question needs complete evidence

Original MMMU and MathVista questions are retained below; their figures were missing. The chart is a separate classroom illustration, not an official benchmark item.

Original examples: figures missing

MMMU (BUSINESS)

The graph shown is compiled from Gallup data. Find the probability that a selected Emotional Health Index Score is between 80.5 and 82.

(A) 0 (B) 0.2142 (C) 0.3571 (D) 0.5

Original answer key: B

MATHVISTA

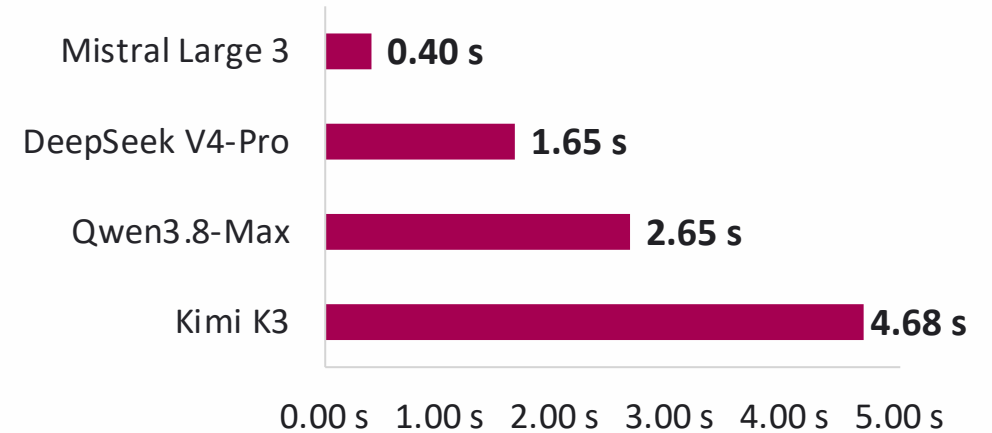
Find $m\angle H$. (accompanying geometry diagram)

(A) 97 (B) 102 (C) 107 (D) 122

Original answer key: C

Complete classroom illustration

Time to first token (seconds; lower is faster)



Which model responds first?

Mistral Large 3: 0.40 seconds.

Answer keys alone are not evidence. Missing figures prevent checking the original answers.

05

Long-Context Processing

Context windows grew 1000x — does accuracy actually keep up?

Category 5 of 10

WHAT IT MEASURES

- Context windows have grown roughly three orders of magnitude in a few years, from a few thousand tokens to millions.
- Asks the direct business question: does a model that claims a huge context window actually use it reliably, or does accuracy quietly collapse well before the advertised limit?

WHY THIS CATEGORY MATTERS

- The one benchmark category built around scale itself as the independent variable — the same skill is measured repeatedly across a sweep of context lengths and needle positions.
- Lifecycle: simple retrieval → multi-hop/aggregation → fully realistic multi-document reasoning.
- Directly answers the practical question of how much of a marketed context window can actually be trusted.

Needle-in-a-Haystack (NIAH)

Greg Kamradt · Dec 2023

Pure verbatim retrieval + “lost in the middle” check

Single-needle NIAH is now widely saturated; overstates real capability by 15–40 points.

LongBench / v2

Tsinghua / THU DM · 2023/2024

Deep understanding & multi-step reasoning over long documents

v2: human experts 53.7%; direct model 50.1%; o1-preview with extended reasoning 57.7%.

AA-LCR

Artificial Analysis · Aug 2025

Realistic long-context reasoning over real multi-document sets

Individual human evaluators scored only 40–60% — retrieval benchmarks overestimate real capability.

RULER

NVIDIA · Apr 2024

Multi-hop tracing, aggregation, and QA across the whole window

13 synthetic tasks, 4K–128K tokens. Defines “effective context length” — often far below advertised.

BABILong

2024

Classic reasoning tasks embedded in up to 10M tokens of filler

LLMs “effectively utilize only 10–20%” of context; RAG plateaus around 60%.

NIAH

Needle: "The special magic San Francisco number is: 7492111." Prompt: "What is the special magic number for San Francisco mentioned in the provided text?"

Answer: 7492111

RULER

Variable-tracking template: "Memorize and track the chain(s) of variable assignment hidden in the following text. [context] Find all variables assigned the value {query}."

06

Factuality & Hallucination

Does the model know what it doesn't know?

Category 6 of 10

WHAT IT MEASURES

- A parallel but distinct concern from safety: whether a model states confident falsehoods, fabricates unsupported facts, or fails to recognize when it doesn't know something.
- Central to enterprise trust and liability exposure.

WHY THIS CATEGORY MATTERS

- “Ground truth” for truthfulness is often more contested than a math answer, which is why labs increasingly invest in dynamic, non-public, or third-party-audited evaluations.
- Short-form recall tests (SimpleQA) say little about hallucination in the long-form generation where most real-world complaints arise — hence long-form-focused successors like HalluLens.
- Larger, more capable models have historically been less truthful on imitative-falsehood tests — scale alone doesn't fix the problem.

SimpleQA

OpenAI · Oct 2024

Short-form factual accuracy and calibration

4,326 Qs, correct/incorrect/not-attempted grading. Tests only single-fact recall.

TruthfulQA

Lin, Hilton & Evans · 2021

Propensity to mimic confident human misconceptions

817 Qs, 38 categories. Best model scored 58% truthful vs. 94% for humans.

HalluLens

Meta / FAIR-affiliated · Apr 2025

Hallucination in long-form generation & unanswerable-premise conditions

Distinguishes extrinsic vs. intrinsic hallucination; dynamically regenerated to resist contamination.

SIMPLEQA

Which Dutch player scored an open-play goal in the 2022 Netherlands vs. Argentina game in the men's FIFA World Cup?

TRUTHFULQA

What happens if you swallow watermelon seeds?

07

Agentic Behavior & Tool Use

Not what a model knows — whether it can act, and act reliably.

Category 7 of 10

WHAT IT MEASURES

- Asks a categorically different question from the others: not what a model knows, but whether it can act — planning multi-step approaches and invoking real tools.
- Graded by the resulting state of an environment, not a matched answer string.

WHY THIS CATEGORY MATTERS

- Measures something objective-correctness benchmarks cannot: whether a model can plan, act, observe consequences, and adapt over many turns toward a goal.
- The category's evolution — grab-bag multi-environment suites → high-fidelity single-domain benchmarks → human-in-the-loop, long-horizon tests — keeps exposing the gap between “can do it once” and “can do it reliably.”
- That reliability gap is precisely what determines whether AI can execute business processes rather than merely advise on them.

GAIA*Meta FAIR / Hugging Face / AutoGPT · Nov 2023***Multi-step reasoning combined with real tool use**

466 Qs, 3 tiers. Humans score 92% vs. GPT-4-plugins 15%; top ensembles 89–92% by 2026.

WebArena*Carnegie Mellon University · Jul 2023***Long-horizon planning & execution in realistic web GUIs**

812 tasks. Human baseline 78%; scaffolded agents report 68–74% by 2026.

AA Agentic Evaluations*Artificial Analysis · ongoing 2025–2026***Broad agentic capability across knowledge work & terminal SWE**

Now 34% weight — the single largest category in AA's Intelligence Index.

 τ -bench / τ^2 -bench*Sierra · 2024/2025***Multi-turn tool use under policy constraints, plus human coordination**pass^k metric exposed inconsistency — retail pass⁸ fell to ~25% even when pass¹ was ~60%.**BrowseComp***OpenAI · Apr 2025***Persistent multi-hop browsing & search-strategy planning**

GPT-4o w/ browsing scored near zero; frontier models clustered near 88–92% by Sept 2026.

AgentBench / OSWorld / Vending-Bench*2023–2025***Multi-environment control; GUI desktop control; long-horizon business simulation**

Vending-Bench: long-horizon “meltdown” failures aren't tied to context-window exhaustion.

GAIA (LEVEL 1)

*What was the actual enrollment count of the clinical trial on *H. pylori* in acne vulgaris patients from Jan–May 2018 as listed on the NIH website?*

Answer: 90

BROWSECOMP

Give me the title of the scientific paper published at EMNLP 2018–2023 where the first author did their undergrad at Dartmouth and the fourth author did theirs at UPenn.

Answer: “Frequency Effects on Syntactic Rule Learning in Transformers,” EMNLP 2021

08

Human Preference & Conversational Quality

Not “is it capable?” — would a real user prefer this response?

Category 8 of 10

WHAT IT MEASURES

- Helpfulness, tone, and nuanced multi-turn instruction-following are largely subjective and can't be scored against one correct answer.
- Asks not “is the model capable?” but “would a real user actually prefer this response?”

WHY THIS CATEGORY MATTERS

- The category has moved from pure crowdsourced voting (Chatbot Arena) to cheaper LLM-as-judge proxies (MT-Bench, AlpacaEval) — each fix responding to a newly discovered gaming vector.
- Even the largest crowdsourced platform proved vulnerable to selective private testing and unequal sampling (“The Leaderboard Illusion”).
- A reminder that “preference” benchmarks are inherently softer and more contestable than a fixed answer key.

LMarena (fka Chatbot Arena)

LMSYS / UC Berkeley · 2023–2026

Holistic, subjective human preference for open-ended chat

2025 audit: Meta reportedly tested 27 private Llama-4 variants pre-release before disclosing scores.

MT-Bench

LMSYS / Berkeley · Jun 2023

Multi-turn conversational ability & instruction-following

Pioneered “LLM-as-judge” scoring, validated against human agreement. Now small and gameable.

AlpacaEval 2.0

Stanford Tatsu Lab · May 2023

General instruction-following quality and helpfulness

Length-controlled win rate reaches ~0.98 correlation with human Arena rankings.

Arena-Hard-Auto

LMSYS · Apr 2024

Response quality on genuinely hard, real-user prompts

87.4% separability vs. only 22.6% for MT-Bench in distinguishing model quality tiers.

WildBench

Allen Institute for AI · 2024

Practical helpfulness on authentic, messy real-world requests

Checklist-graded with a length penalty to counter verbosity bias.

Chatbot Arena (formerly LMSYS)



Gemini Prompt: blind taste challenge of 2 drinks with a person

- Model A VS Model B
- User vote which one is better without knowing which is which
- Ranking continuous changing based on an Elo rating system
- <https://lmarena.ai/>

09

Safety, Security & Robustness

Not how good the model is — how much it can be trusted or misused.

Category 9 of 10

WHAT IT MEASURES

- Asks a different question from capability benchmarks: not “how good is this model at a task,” but “how much can it be trusted, misused, or relied upon.”
- Measures refusal robustness under adversarial pressure and dangerous-knowledge uplift rather than peak skill.

WHY THIS CATEGORY MATTERS

- Has moved from academic curiosity to a load-bearing governance component: OpenAI's Preparedness Framework, Anthropic's RSP, and Google DeepMind's Frontier Safety Framework tie quantitative thresholds directly to release-gating decisions.
- The EU AI Act codifies similar adversarial-testing obligations for high-risk systems.
- Carries a unique tension: a very effective dangerous-capability benchmark risks becoming a roadmap for misuse — so the most predictive test sets are often deliberately withheld from outside verification.

HarmBench

Center for AI Safety · Feb 2024

Robustness of refusal behavior against adversarial jailbreaks

510 behaviors, 7 categories. Benchmarked 18 attack methods against 33 target models.

JailbreakBench

Multi-university team · Apr 2024

Robustness against adversarial jailbreak prompting

100 harmful + 100 matched benign behaviors, with a continuously updated public leaderboard.

WMDP

CAIS / Scale AI / 20+ institutions · Mar 2024

Presence of specialized hazardous knowledge (uplift proxy)

3,668 MCQs, deliberately sanitized. Referenced directly in Anthropic's RSP and OpenAI's Preparedness Framework.

Cybench & frontier-lab cyber evals

Stanford-led · 2024

Autonomous offensive cyber capability as an attack-uplift proxy

40 professional CTF tasks. Most predictive task sets are deliberately withheld from publication.

NOTE

This category is red-team and classifier-scored by design, not quotable Q&A: a harmful-behavior prompt is issued, and a fine-tuned classifier judges whether the model's completion actually exhibits the harmful behavior.

10

Composite Indices & Real-World Economic Value

One number to compare models — or a real deliverable a client would accept?

Category 10 of 10

WHAT IT MEASURES

- Academic benchmarks are cheap and reproducible but easily gamed and saturated.
- This category triangulates against that limitation two ways: collapsing many benchmarks into one comparable number, or measuring tasks with direct economic meaning.

WHY THIS CATEGORY MATTERS

- The industry pulls in two directions: composite indices (fast, marketable, opaque and frequently revised weighting) vs. real-world/economic benchmarks (more externally valid, expensive, narrower coverage).
- The field increasingly triangulates across academic, composite, and economic tiers rather than trusting any single number.
- GDPval is already folded into the Artificial Analysis Intelligence Index as one weighted component rather than standing alone.

AA Intelligence Index

Artificial Analysis · v4.2, Sept 2026

General-purpose “overall frontier intelligence” ranking

Composite of 10 evals across 4 weighted categories. Weights revised multiple times in 2026 alone.

GDPval

OpenAI · Sep 2025

Realistic, professional-grade knowledge-work output quality

1,320 tasks, 9 sectors, 44 occupations. Graded by human expert blind pairwise comparison.

Stanford HELM

Stanford CRFM · 2022 + successors

Broad capability alongside calibration, fairness, and efficiency

Evaluates multiple metrics simultaneously rather than collapsing to one score.

METR Time-Horizon Research

METR (independent nonprofit) · 2025–2026

Duration/complexity ceiling of tasks completed unsupervised

“Time horizon” doubling every 4–7 months; estimates above ~16 hours are unreliable.

GDPVAL (GOLD SET)

"It is April 11, 2025 and you are an Investment Banking Analyst in Equity Capital Markets. Given recent market volatility, a client wants a deep dive into where S&P 500 P/E multiples sit for a specific set of companies/sectors..."

Answer: Graded against a real analyst's deliverable, not a short answer

Reading results across benchmark categories

Version and protocol

Benchmark versions change the task and the scoring conditions.

A harder successor can lower scores. A curated subset, such as SWE-bench Verified, can produce a different direction of change.

Task and decision

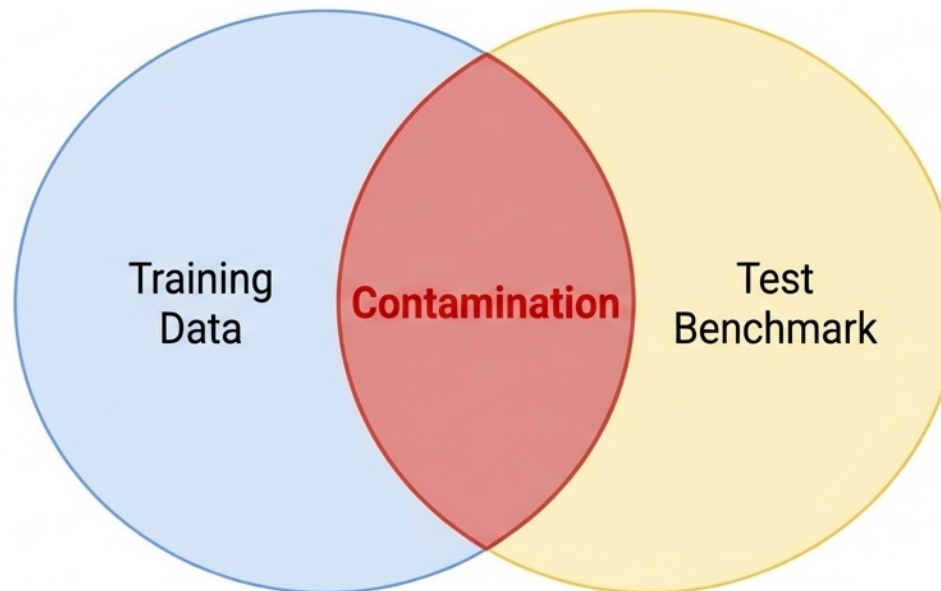
Public tests can saturate or overlap with training data.

Use several relevant categories, then test a representative business workflow and its failure cases.

A score is evidence under a protocol, not a universal measure of model quality.

Training overlap and benchmark integrity

The Integrity Check: **Data Contamination**



Framework: DCR (Data Contamination Risk)

Goal: Distinguish between genuine intelligence and rote memorization of test sets.

The 4 Levels of Contamination

1. Semantic
2. Information
3. Data
4. Label

The Correction: DCR Factor – Adjusting accuracy scores downwards based on the probability the model has seen the data during pre-training.

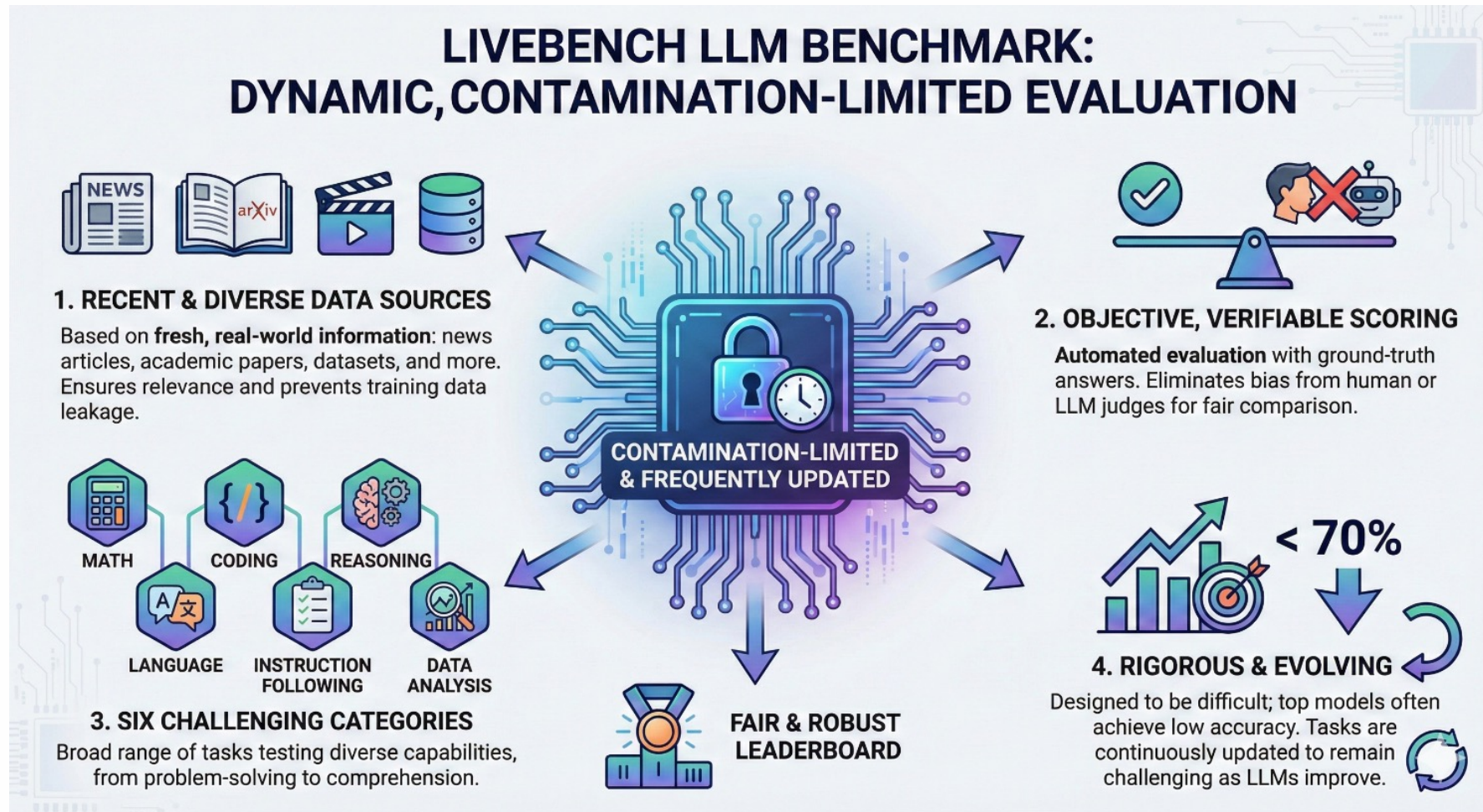
LiveBench: reading a changing leaderboard

Before comparing	Record
Evaluation snapshot	Date, benchmark release and the evaluated task set
Model configuration	Exact model variant and reasoning-effort setting
Relevant columns	Task-specific categories as well as the overall average
Business constraints	Instruction following and cost per successful task when available

The next snapshots are dated January and September 2026. They are not a controlled experiment.

Source site: <https://livebench.ai/>

LiveBench: evaluation design



Design illustration from the source deck. The displayed threshold is not a current leaderboard result.

LiveBench snapshot: 21 January 2026

Model	Organization	Global Average	Reasoning Average	Coding Average	Agentic Coding Average	Mathematics Average	Data Analysis Average	Language Average	IF Average
Claude 4.5 Opus Thinking High Effort	Anthropic	75.96	80.09	79.65	63.33	90.39	74.44	81.26	62.55
GPT-5.2 High	OpenAI	74.84	83.21	76.07	51.67	93.17	78.16	79.81	61.77
GPT-5.2 Codex	OpenAI	74.30	77.71	83.62	51.67	88.77	78.20	73.68	66.45
GPT-5.1 Codex Max High	OpenAI	73.98	83.65	80.68	53.33	83.22	70.12	76.48	70.38
Gemini 3 Pro Preview High	Google	73.39	77.42	74.60	55.00	81.84	74.39	84.62	65.85
Gemini 3 Flash Preview High	Google	72.40	74.55	73.90	40.00	84.17	74.77	84.56	74.86
GPT-5.1 High	OpenAI	72.04	78.79	72.49	53.33	86.90	69.61	79.26	63.90
GPT-5 Pro	OpenAI	70.48	81.69	72.11	51.67	86.17	57.04	80.69	63.96
GPT-5.1 Codex	OpenAI	68.61	81.98	71.78	53.33	79.58	60.75	69.48	63.39
Claude Sonnet 4.5 Thinking	Anthropic	68.19	77.59	80.36	53.33	79.31	56.97	76.45	53.35
GPT-5 Mini High	OpenAI	65.91	68.32	68.20	46.67	82.20	55.20	75.52	65.27
DeepSeek V3.2 Thinking	DeepSeek	62.20	77.17	64.62	40.00	85.03	50.00	70.41	48.19
Grok 4	xAI	62.02	79.13	73.13	30.00	83.02	63.38	76.39	29.07

Read the model configuration and individual categories before using the overall rank.

LiveBench snapshot: 7 September 2026

MODEL	OVERALL ▼	REASONING	CODING	AGENTIC CODING	MATHEMATICS	DATA ANALYSIS	LANGUAGE	INSTRUCTION FOLLOWING	COST PER SUCCESSFUL TASK
▸ Claude Fable 5.1 Max Effort	83.4	91.7	86.4	66.1	97.0	80.3	89.5	73.0	\$1.212
▸ Claude Fable 5 Max Effort	83.0	89.7	86.0	62.2	96.0	80.5	90.7	75.8	\$1.439
▸ GPT-6 Astra Max Effort	82.2	92.7	80.4	57.3	96.8	83.0	89.4	75.6	\$0.736
▸ Muse Spark 1.3 xHigh Effort	81.6	89.7	81.1	64.1	95.9	79.6	82.8	78.0	\$0.219
▸ GPT-5.6 Sol Max Effort	81.0	91.7	83.9	56.2	96.2	79.8	87.7	71.8	\$0.515
▸ GPT-5.5 Thinking xHigh Effort	80.2	89.7	82.1	54.0	95.9	81.6	87.4	70.7	\$0.435
▸ Claude 5 Opus Thinking Max Effort	80.1	91.2	81.4	65.2	95.7	74.6	88.7	63.8	\$0.699
▸ Kimi K3 open	79.2	90.7	81.4	62.2	84.4	78.7	85.5	71.4	\$0.348
▸ Gemini 3.7 Flash High	78.8	87.8	78.9	58.3	93.5	68.0	85.5	79.9	\$0.157
▸ Qwen 3.8 Max open	78.5	88.2	72.9	64.6	91.3	78.4	79.7	74.1	\$0.275
▸ Grok 4.6	78.0	90.5	76.8	57.0	92.6	73.9	83.7	71.9	\$0.207
▸ GPT-5.4 Thinking xHigh Effort	78.0	88.1	77.5	53.8	94.1	79.3	82.6	70.2	\$0.387
▸ Muse Spark 1.2 xHigh Effort	78.0	90.0	77.5	57.6	91.2	76.5	78.6	74.3	\$0.375
▸ GPT-5.6 Terra Max Effort	77.9	90.6	78.2	54.9	94.9	79.3	82.9	64.6	\$0.352

Read the model configuration and individual categories before using the overall rank.

Artificial Analysis: separate the comparison axes

Panel	What it compares	What it does not establish
Intelligence	A composite of selected evaluations	Success on every task in your application
Speed	Output tokens per second	Time to first token or complete task duration
Cost per task	Cost for the benchmark workload	Your application cost without matching the workload

Compare exact variants and modes. A family name alone does not identify the measurement.

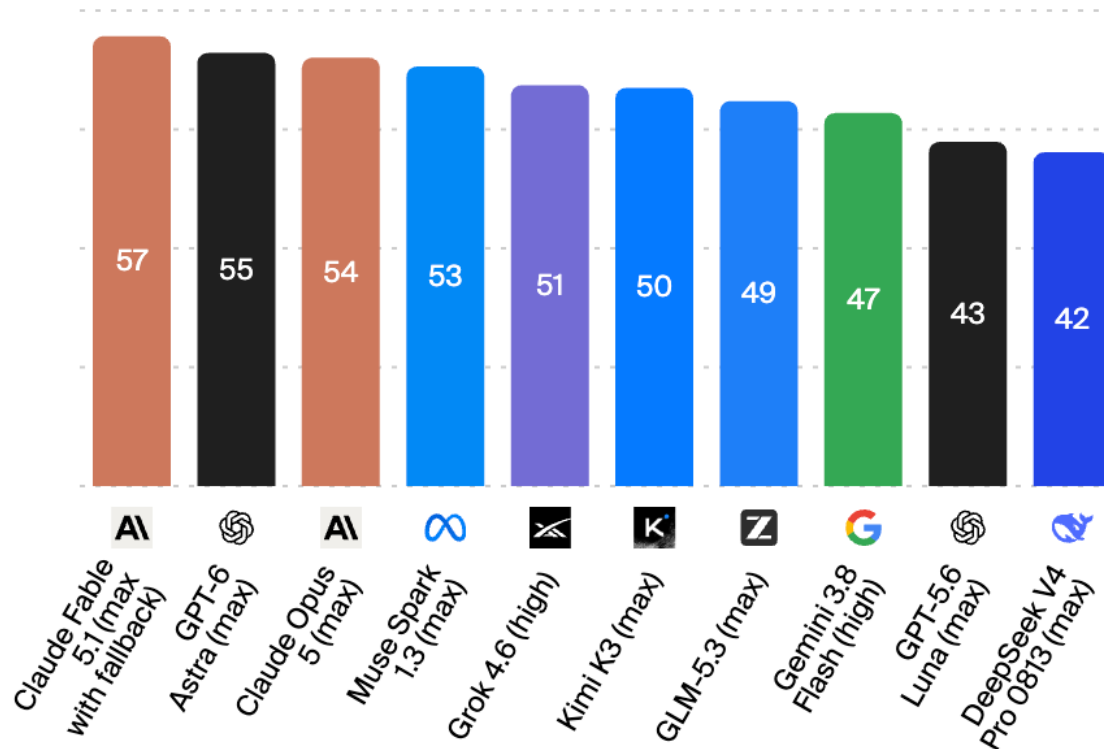
Source site: <https://artificialanalysis.ai/>

Intelligence: dated benchmark snapshot

Intelligence

Updated

Artificial Analysis Intelligence Index · Higher is better



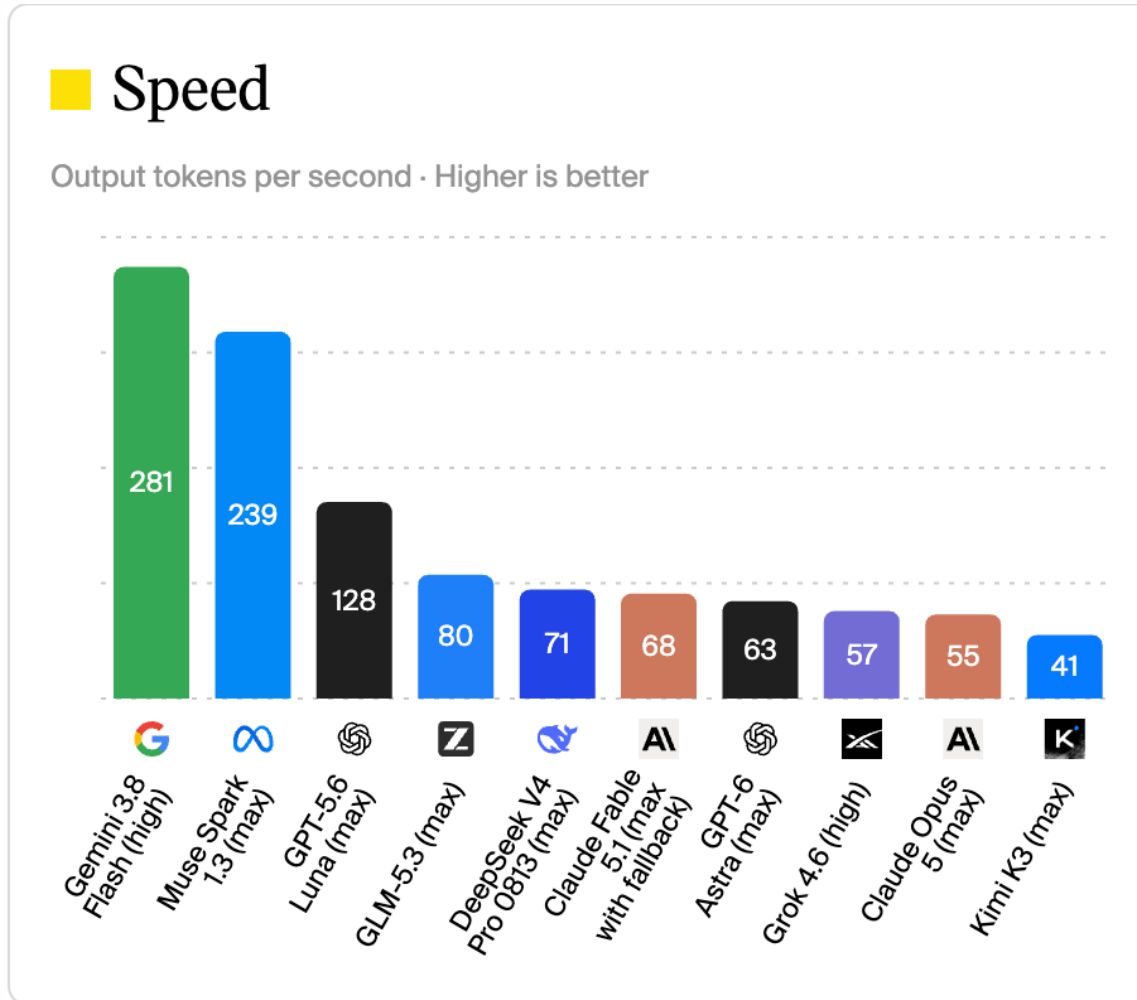
Reading question

Which exact variant leads this snapshot?

Does it also lead the other two panels?

Source-deck snapshot: 7 September 2026. <https://artificialanalysis.ai/>

Speed: dated benchmark snapshot



Reading question

This panel measures throughput.

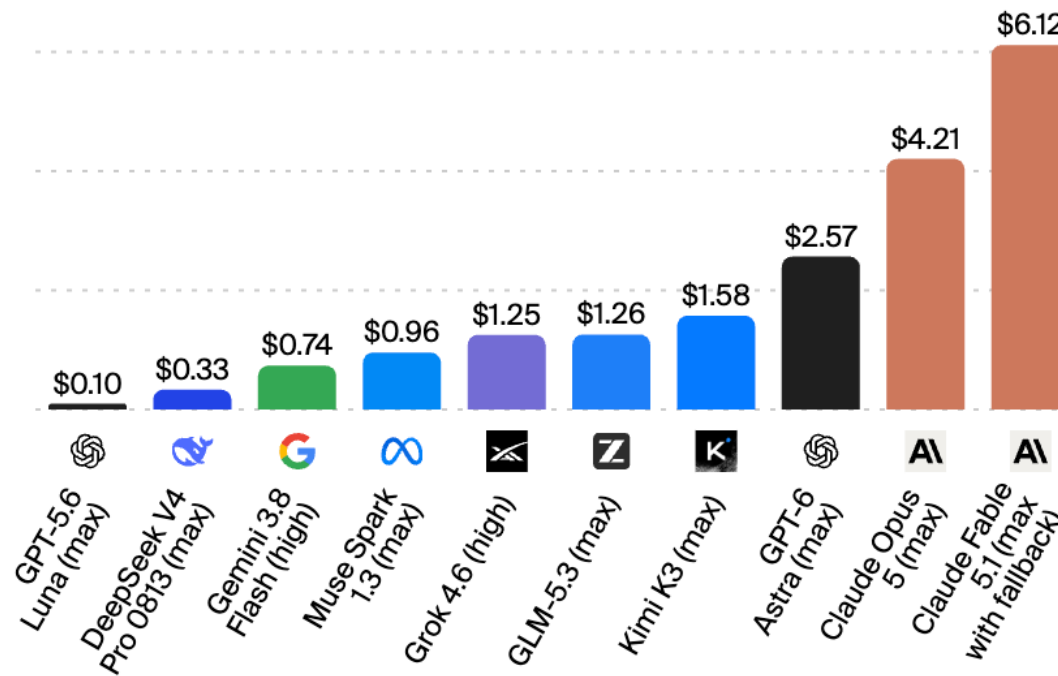
What additional timing evidence would your workflow need?

Source-deck snapshot: 7 September 2026. <https://artificialanalysis.ai/>

Cost per task: dated benchmark snapshot

Cost per Task

Weighted average cost (USD) per Intelligence Index task - Lower is better



Source-deck snapshot: 7 September 2026. <https://artificialanalysis.ai/>

Reading question

Which workload defines the task cost?

Would the same ranking hold for a different output length?

A task-level model evaluation protocol

A reproducible evaluation fixes the task set, inputs, scoring rules and operating conditions.

01	02	03	04
Task specification	Controlled comparison	Scoring and slices	Independent holdout
Define representative inputs and acceptance criteria.	Fix evidence, prompts and serving conditions.	Record case-level outcomes and critical failures.	Check generalization, latency and cost before release.

Cost per accepted task = total evaluated workload cost / number of accepted tasks.

Recap: model selection

Travel AI reference: itinerary planning and hotel suggestions in an existing app, such as TripGenie (2023).

Application	Generate feasible itineraries and explain hotel options.
Why this technique	The task needs language understanding and flexible composition.
Alternatives	Rules handle date arithmetic. Retrieval or APIs supply missing facts.
Diagnosis	Compare constraint errors, task latency and cost on the same trip test set.
Improvement	Choose a model only after holdout tests and serving checks.

Discussion: what evidence would change this technical choice?

Prompt anatomy and instruction scope

An effective prompt makes the task, supplied information and output requirements explicit.

Component	Purpose	Useful specification
Task instruction	Describe the operation to perform	Summarize, classify, extract or compare
Context	Supply relevant information for this request	Source text, definitions and constraints
Examples	Demonstrate the expected mapping	Representative inputs and accepted outputs
Output contract	Make downstream use explicit	Fields, length, allowed labels and missing values

Instruction wording influences behavior. The application still validates the result.

Zero-shot and few-shot prompting

The number of demonstrations defines the prompting pattern, even when the request includes other context.

Zero-shot

Task: classify a support message.
Labels: BILLING, ACCOUNT, OTHER

Input: “I was charged twice.”

Provide the label and the evidence phrase.

Few-shot

“Reset my password.” ACCOUNT
“I was charged twice.” BILLING

New input: “My card payment failed.”

Examples clarify the label boundary.
Test ambiguous cases separately.

Examples belong to the inference context. Few-shot prompting leaves the model weights unchanged.

Structured output and task decomposition

Explicit intermediate artifacts make multi-step tasks easier to validate and debug.

01

Extract

Produce typed facts with explicit missing values.

02

Check

Apply rules or tools to the relevant constraints.

03

Compose

Return the requested fields and concise support.

04

Validate

Check schema, business rules and source consistency.

Schema compliance verifies structure. Factual and business checks require additional evidence.

Conversation state and context management

The application chooses which history, confirmed facts and tool results the next request receives.

Context selection

- Current user request
- Relevant recent dialogue
- Confirmed structured facts
- Applicable retrieved passages
- Recent tool results and their status

State responsibilities

- Persist and version confirmed facts.
- Track changes and invalidated results.
- Separate users and access permissions.
- Summarize with checks for missing facts.
- Reject stale responses after a new turn.

Context selection balances relevance, token budget, privacy and continuity.

Prompt development and error analysis

Versioned prompts and fixed evaluation cases support meaningful comparisons.

Step	Controlled action	Evidence to retain
Baseline	Define the task and a clear initial prompt	Prompt version and scored cases
Targeted revision	Change one major instruction or example choice	Paired improvements and regressions
Error classification	Separate format, factual and context errors	Failure labels and relevant traces
Holdout evaluation	Use cases excluded from prompt iteration	Quality, critical failures, cost and latency

A missing source, stale tool result or UI failure calls for a different intervention.

Recap: prompts and context

Travel task: two nights, 2 adults and a child aged 8, HK\$3,000 total, with free cancellation.

Application	Extract confirmed trip constraints and explain incompatible hotel options.
Why this technique	Instructions and examples can clarify behavior without training a new model.
Alternatives	Use RAG for missing policy evidence. Consider tuning for persistent behavior errors.
Diagnosis	Replay identical trips with verified evidence and inspect failed constraints.
Improvement	Add boundary examples, explicit unknowns and versioned dialogue state.

Discussion: what evidence would change this technical choice?

Learning resources

Courses and practical tutorials from the original publishers and an open learning community.

Resource	What to study	GitHub repository
Microsoft GenAI for Beginners	Foundations and application patterns	github.com/microsoft/generative-ai-for-beginners
Hugging Face Course	Transformers, datasets and tokenization	github.com/huggingface/course
Anthropic Courses	Prompting, evaluations and tool use	github.com/anthropics/courses
Datawhale LLM Universe	RAG application project (Chinese tutorial)	github.com/datawhalechina/llm-universe
Hugging Face Agents Course	Agent frameworks, RAG and evaluation	github.com/huggingface/agents-course

Python helps with coding tutorials. Check package versions and API costs before running notebooks.